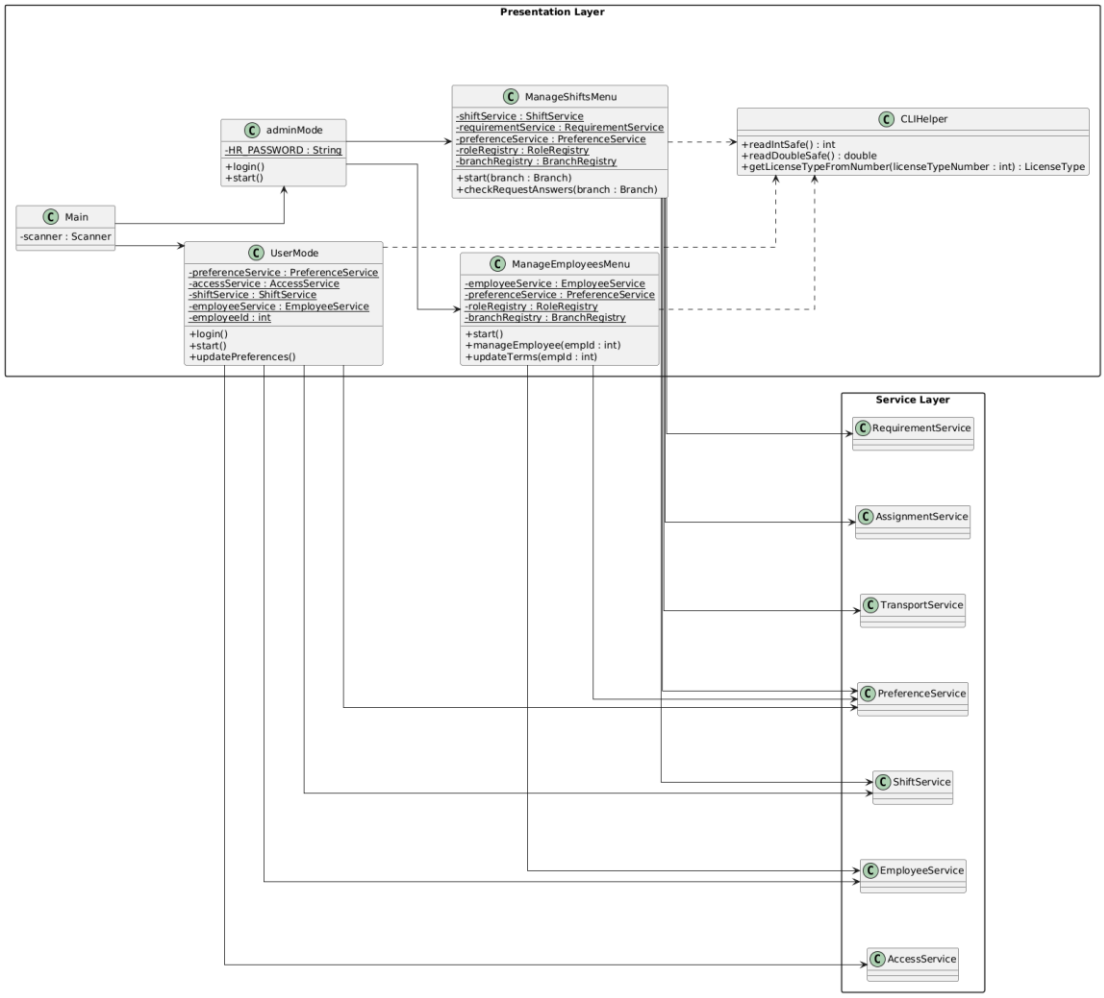
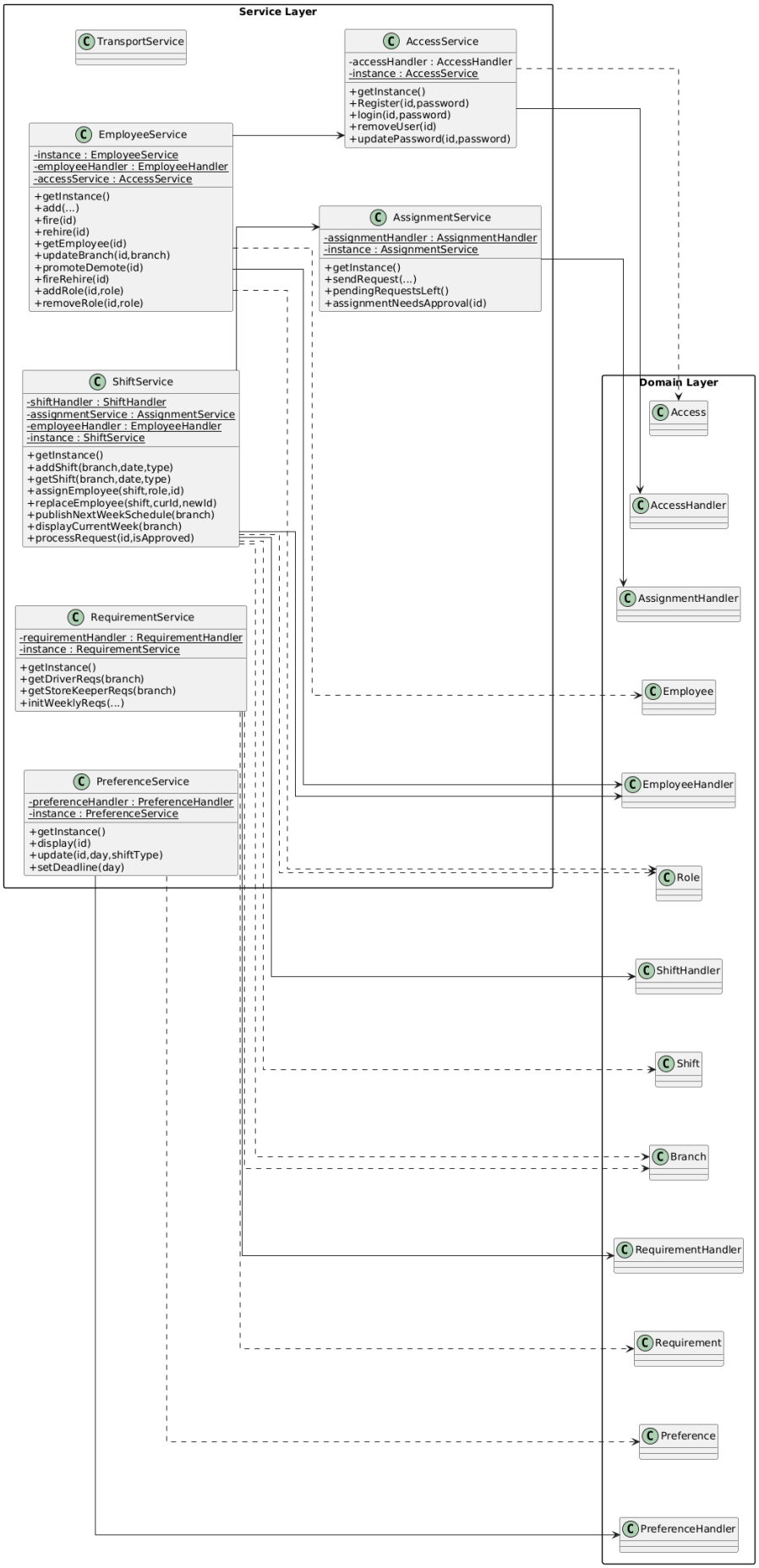
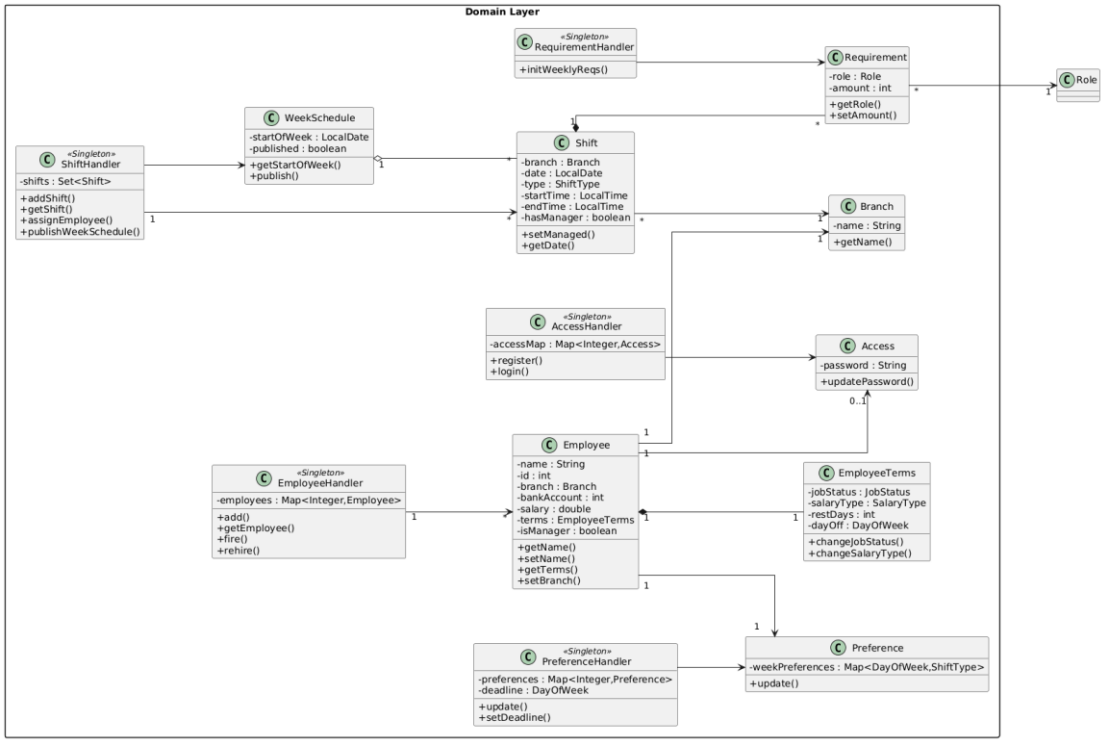
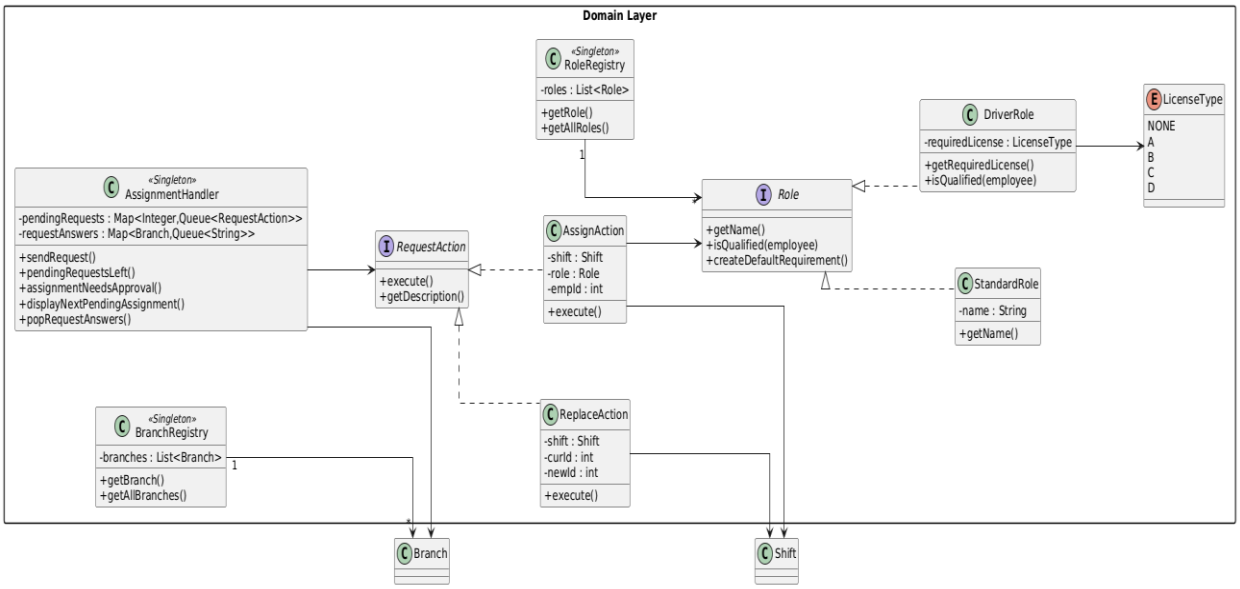
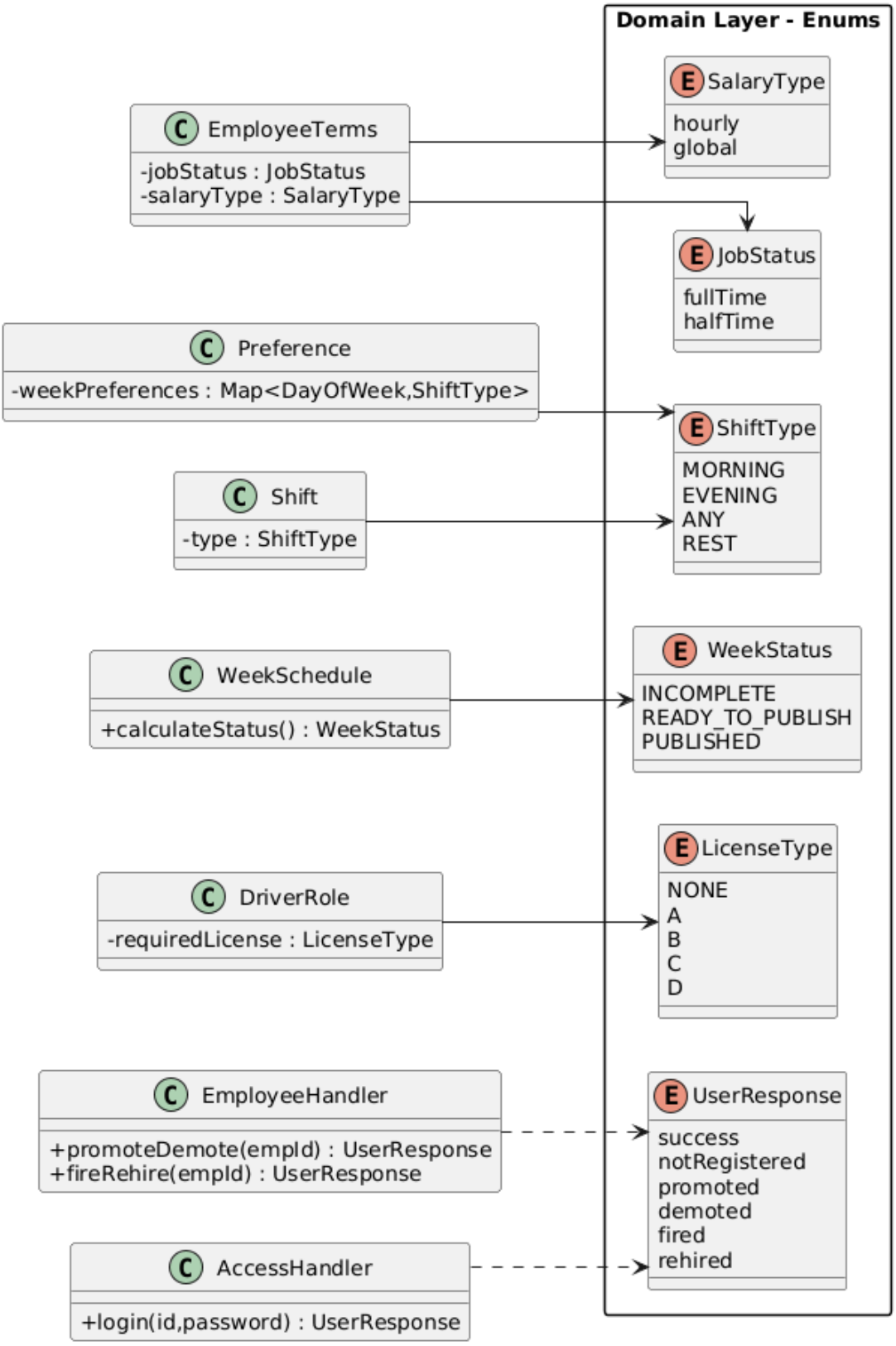
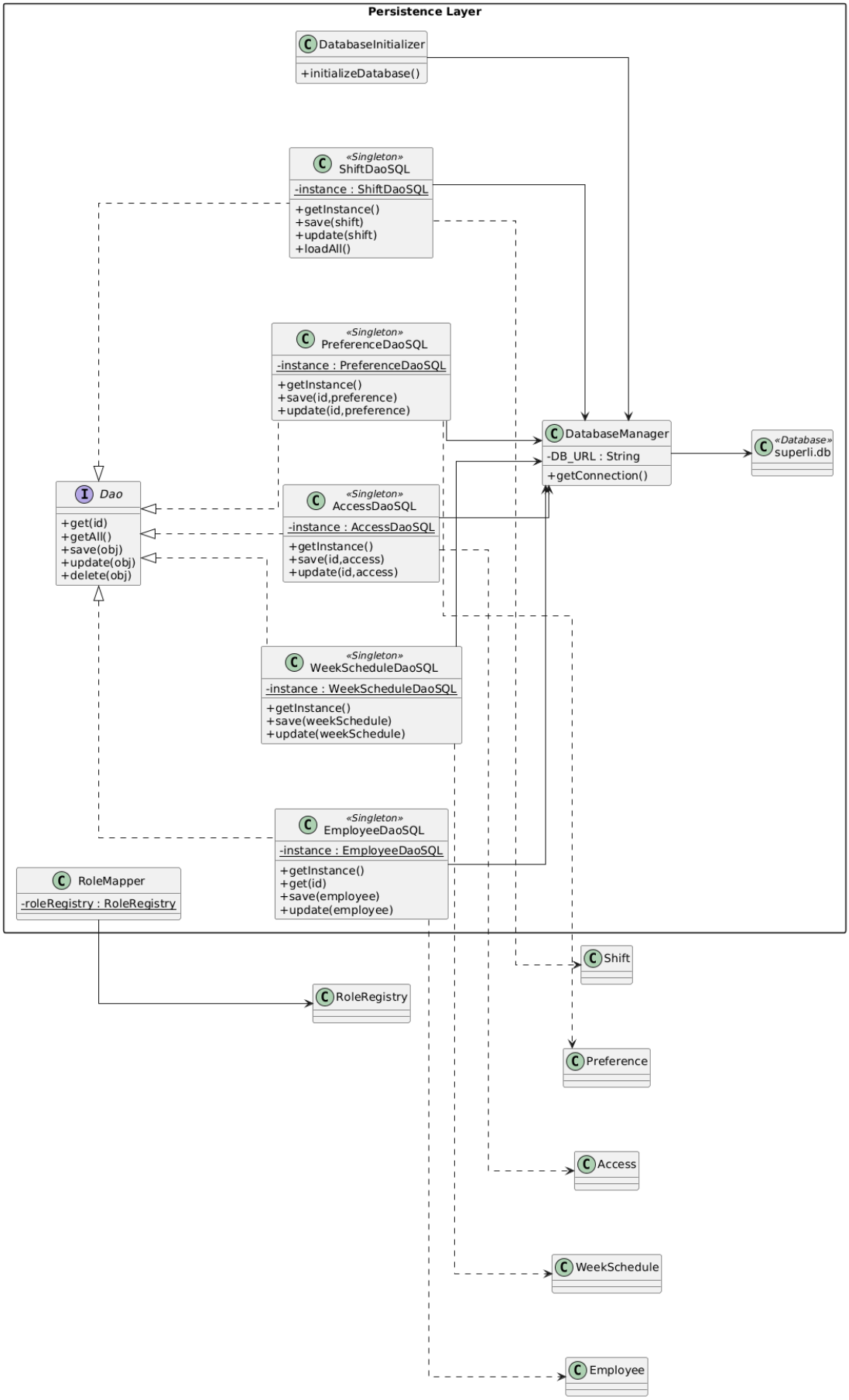

Presentation Layer
ManageShiftsMenu
-shiftService : ShiftService
-requirementService : RequirementService
-preferenceService : PreferenceService
-roleRegistry : RoleRegistry
-branchRegistry : BranchRegistry
+start(branch : Branch)
+checkRequestAnswers(branch : Branch)
CLIHelper
+readIntSafe() : int
+readDoubleSafe() : double
+getLicenseTypeFromNumber(licenseTypeNumber : int) : LicenseType
adminMode
-HR_PASSWORD : String
+login()
+start()
Main
-scanner : Scanner
UserMode
-preferenceService : PreferenceService
-accessService : AccessService
-shiftService : ShiftService
-employeeService : EmployeeService
-employeeId : int
+login()
+start()
+updatePreferences()
ManageEmployeesMenu
-employeeService : EmployeeService
-preferenceService : PreferenceService
-roleRegistry : RoleRegistry
-branchRegistry : BranchRegistry
+start()
+manageEmployee(empId : int)
+updateTerms(empId : int)
Service Layer
RequirementService
AssignmentService
TransportService
PreferenceService
ShiftService
EmployeeService
AccessService

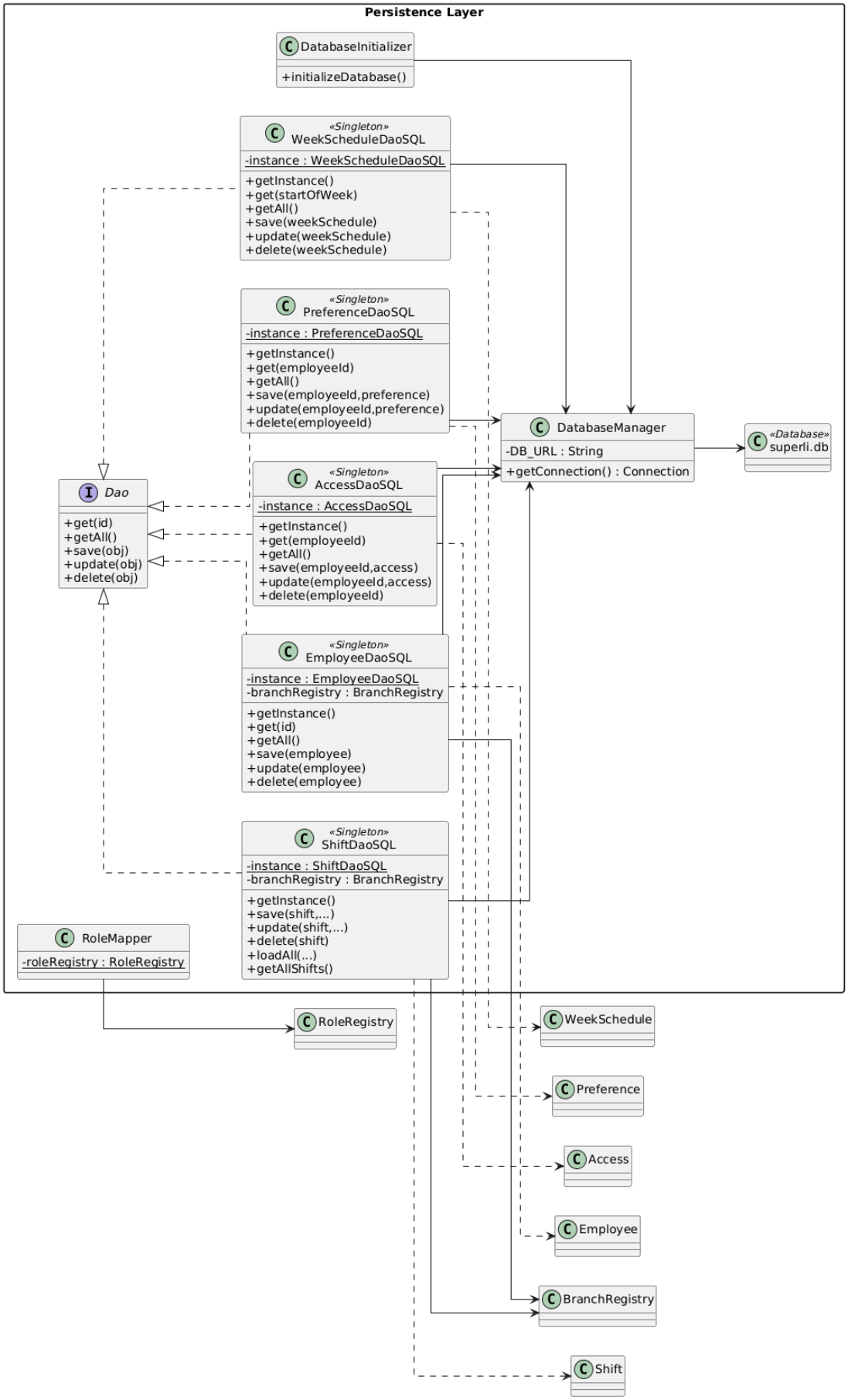

Service Layer
TransportService
AccessService
-accessHandler : AccessHandler
-instance : AccessService
+getInstance()
+Register(id,password)
+login(id,password)
+removeUser(id)
+updatePassword(id,password)
EmployeeService
-instance : EmployeeService
-employeeHandler : EmployeeHandler
-accessService : AccessService
+getInstance()
+add(...)
+fire(id)
+rehire(id)
+getEmployee(id)
+updateBranch(id,branch)
+promoteDemote(id)
+fireRehire(id)
+addRole(id,role)
+removeRole(id,role)
AssignmentService
-assignmentHandler : AssignmentHandler
-instance : AssignmentService
+getInstance()
+sendRequest(...)
+pendingRequestsLeft()
+assignmentNeedsApproval(id)
ShiftService
-shiftHandler : ShiftHandler
-assignmentService : AssignmentService
-employeeHandler : EmployeeHandler
-instance : ShiftService
+getInstance()
+addShift(branch,date,type)
+getShift(branch,date,type)
+assignEmployee(shift,role,id)
+replaceEmployee(shift,curId,newId)
+publishNextWeekSchedule(branch)
+displayCurrentWeek(branch)
+processRequest(id,isApproved)
RequirementService
-requirementHandler : RequirementHandler
-instance : RequirementService
+getInstance()
+getDriverReqs(branch)
+getStoreKeeperReqs(branch)
+initWeeklyReqs(...)
PreferenceService
-preferenceHandler : PreferenceHandler
-instance : PreferenceService
+getInstance()
+display(id)
+update(id,day,shiftType)
+setDeadline(day)
Domain Layer
Access
AccessHandler
AssignmentHandler
Employee
EmployeeHandler
Role
ShiftHandler
Shift
Branch
RequirementHandler
Requirement
Preference
PreferenceHandler

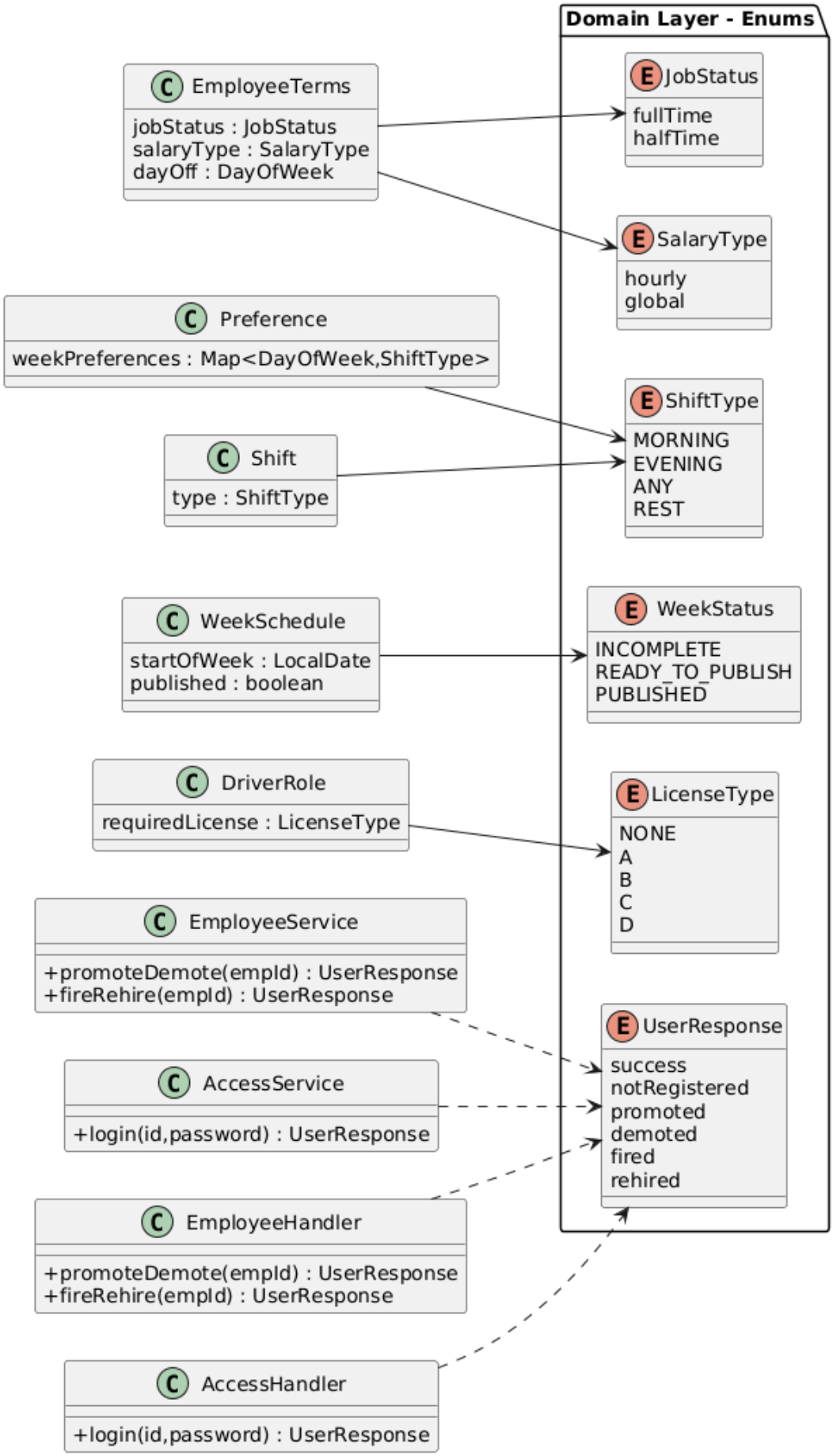

Domain Layer
«Singleton»
RequirementHandler
+initWeeklyReqs()
Requirement
-role : Role
-amount : int
+getRole()
+setAmount()
Role
WeekSchedule
-startOfWeek : LocalDate
-published : boolean
+getStartOfWeek()
+publish()
«Singleton»
ShiftHandler
-shifts : Set<Shift>
+addShift()
+getShift()
+assignEmployee()
+publishWeekSchedule()
Shift
-branch : Branch
-date : LocalDate
-type : ShiftType
-startTime : LocalTime
-endTime : LocalTime
-hasManager : boolean
+setManaged()
+getDate()
Branch
-name : String
+getName()
«Singleton»
AccessHandler
-accessMap : Map<Integer,Access>
+register()
+login()
Access
-password : String
+updatePassword()
0..1
Employee
-name : String
-id : int
-branch : Branch
-bankAccount : int
-salary : double
-terms : EmployeeTerms
-isManager : boolean
+getName()
+setName()
+getTerms()
+setBranch()
«Singleton»
EmployeeHandler
-employees : Map<Integer,Employee>
+add()
+getEmployee()
+fire()
+rehire()
EmployeeTerms
-jobStatus : JobStatus
-salaryType : SalaryType
-restDays : int
-dayOff : DayOfWeek
+changeJobStatus()
+changeSalaryType()
«Singleton»
PreferenceHandler
-preferences : Map<Integer,Preference>
-deadline : DayOfWeek
+update()
+setDeadline()
Preference
-weekPreferences : Map<DayOfWeek,ShiftType>
+update()


Domain Layer
«Singleton»
RoleRegistry
-roles : List<Role>
+getRole()
+getAllRoles()
DriverRole
-requiredLicense : LicenseType
+getRequiredLicense()
+isQualified(employee)
LicenseType
NONE
A
B
C
D
«Singleton»
AssignmentHandler
-pendingRequests : Map<Integer,Queue<RequestAction>>
-requestAnswers : Map<Branch,Queue<String>>
+sendRequest()
+pendingRequestsLeft()
+assignmentNeedsApproval()
+displayNextPendingAssignment()
+popRequestAnswers()
Role
+getName()
+isQualified(employee)
+createDefaultRequirement()
RequestAction
+execute()
+getDescription()
AssignAction
-shift : Shift
-role : Role
-empId : int
+execute()
StandardRole
-name : String
+getName()
«Singleton»
BranchRegistry
-branches : List<Branch>
+getBranch()
+getAllBranches()
ReplaceAction
-shift : Shift
-curId : int
-newId : int
+execute()
Branch
Shift


Domain Layer - Enums
EmployeeTerms
-jobStatus : JobStatus
-salaryType : SalaryType
SalaryType
hourly
global
JobStatus
fullTime
halfTime
Preference
-weekPreferences : Map<DayOfWeek,ShiftType>
ShiftType
MORNING
EVENING
ANY
REST
Shift
-type : ShiftType
WeekSchedule
+calculateStatus() : WeekStatus
WeekStatus
INCOMPLETE
READY_TO_PUBLISH
PUBLISHED
LicenseType
NONE
A
B
C
D
DriverRole
-requiredLicense : LicenseType
EmployeeHandler
+promoteDemote(empId) : UserResponse
+fireRehire(empId) : UserResponse
UserResponse
success
notRegistered
promoted
demoted
fired
rehired
AccessHandler
+login(id,password) : UserResponse


Persistence Layer
DatabaseInitializer
+initializeDatabase()
«Singleton»
ShiftDaoSQL
-instance : ShiftDaoSQL
+getInstance()
+save(shift)
+update(shift)
+loadAll()
«Singleton»
PreferenceDaoSQL
-instance : PreferenceDaoSQL
+getInstance()
+save(id,preference)
+update(id,preference)
DatabaseManager
-DB_URL : String
+getConnection()
«Database»
superli.db
Dao
+get(id)
+getAll()
+save(obj)
+update(obj)
+delete(obj)
«Singleton»
AccessDaoSQL
-instance : AccessDaoSQL
+getInstance()
+save(id,access)
+update(id,access)
«Singleton»
WeekScheduleDaoSQL
-instance : WeekScheduleDaoSQL
+getInstance()
+save(weekSchedule)
+update(weekSchedule)
«Singleton»
EmployeeDaoSQL
-instance : EmployeeDaoSQL
+getInstance()
+get(id)
+save(employee)
+update(employee)
RoleMapper
-roleRegistry : RoleRegistry
Shift
RoleRegistry
Preference
Access
WeekSchedule
Employee


Persistence Layer
DatabaseInitializer
+initializeDatabase()
«Singleton»
WeekScheduleDaoSQL
-instance : WeekScheduleDaoSQL
+getInstance()
+get(startOfWeek)
+getAll()
+save(weekSchedule)
+update(weekSchedule)
+delete(weekSchedule)
«Singleton»
PreferenceDaoSQL
-instance : PreferenceDaoSQL
+getInstance()
+get(employeeId)
+getAll()
+save(employeeId,preference)
+update(employeeId,preference)
+delete(employeeId)
DatabaseManager
-DB_URL : String
+getConnection() : Connection
«Database»
superli.db
Dao
+get(id)
+getAll()
+save(obj)
+update(obj)
+delete(obj)
«Singleton»
AccessDaoSQL
-instance : AccessDaoSQL
+getInstance()
+get(employeeId)
+getAll()
+save(employeeId,access)
+update(employeeId,access)
+delete(employeeId)
«Singleton»
EmployeeDaoSQL
-instance : EmployeeDaoSQL
-branchRegistry : BranchRegistry
+getInstance()
+get(id)
+getAll()
+save(employee)
+update(employee)
+delete(employee)
«Singleton»
ShiftDaoSQL
-instance : ShiftDaoSQL
-branchRegistry : BranchRegistry
+getInstance()
+save(shift,...)
+update(shift,...)
+delete(shift)
+loadAll(...)
+getAllShifts()
RoleMapper
-roleRegistry : RoleRegistry
RoleRegistry
WeekSchedule
Preference
Access
Employee
BranchRegistry
Shift


Domain Layer - Enums
EmployeeTerms
jobStatus : JobStatus
salaryType : SalaryType
dayOff : DayOfWeek
JobStatus
fullTime
halfTime
SalaryType
hourly
global
Preference
weekPreferences : Map<DayOfWeek,ShiftType>
ShiftType
MORNING
EVENING
ANY
REST
Shift
type : ShiftType
WeekSchedule
startOfWeek : LocalDate
published : boolean
WeekStatus
INCOMPLETE
READY_TO_PUBLISH
PUBLISHED
DriverRole
requiredLicense : LicenseType
LicenseType
NONE
A
B
C
D
EmployeeService
+promoteDemote(empId) : UserResponse
+fireRehire(empId) : UserResponse
UserResponse
success
notRegistered
promoted
demoted
fired
rehired
AccessService
+login(id,password) : UserResponse
EmployeeHandler
+promoteDemote(empId) : UserResponse
+fireRehire(empId) : UserResponse
AccessHandler
+login(id,password) : UserResponse